

[5927]-217

B.E. (Mechanical)

CAD/CAM & AUTOMATION

(2015 Pattern) (Semester - I) (402042)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Use of calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) A triangle ABC has its vertices at A(10,10), B(10, 20), C(20, 10) is rotated about Origin 'O' in 90° in CCW direction. Calculate concatenated transformation matrix and determine final position of triangle. [6]

b) Write parametric equations of (i) Circle, (ii) Ellipse. [4]

OR

Q2) a) Differentiate between Orthogonal and Perspective Projections. [6]

b) Explain Cubic spline curve with neat sketch. [4]

Q3) a) Explain Coon's patch surface with neat sketch. [5]

b) Distinguish between 'h' & 'p' formulations. [5]

OR

Q4) An axial stepped bar as shown in Figure 1, is axially loaded by force $F_1 = 10kN$ and $F_2 = 10kN$. Assuming $E_1 = E_2 = 200N/mm^2$, $A_1 = 200mm^2$, $A_2 = 100mm^2$ and $L_1 = L_2 = 200mm$, determine deflection, stresses in elements and reactions at nodes. [10]

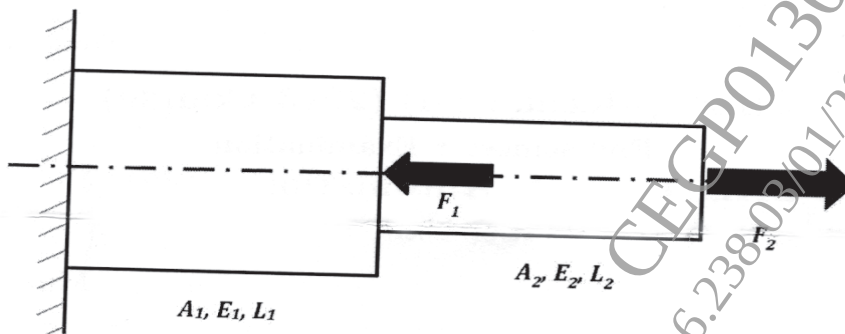
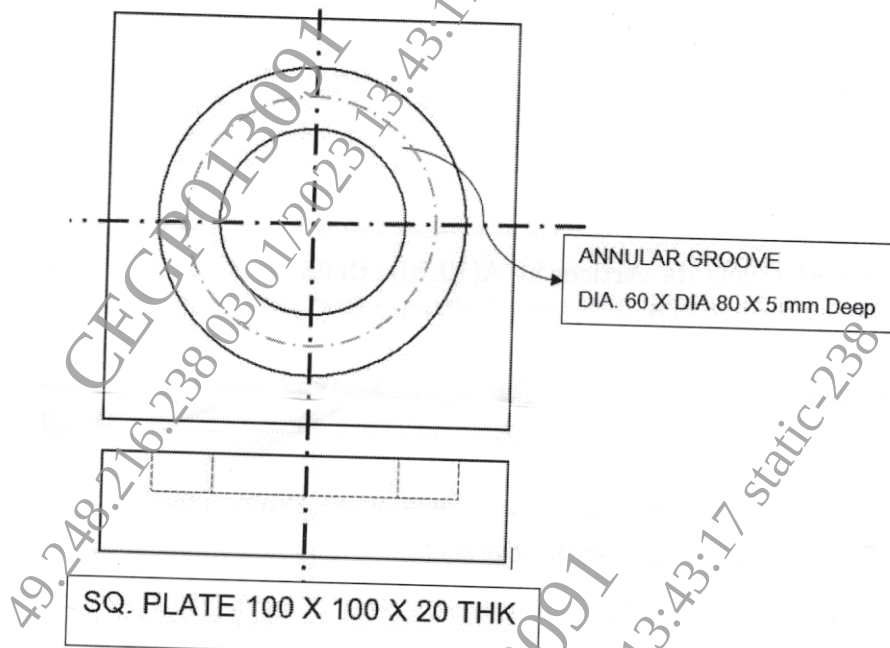


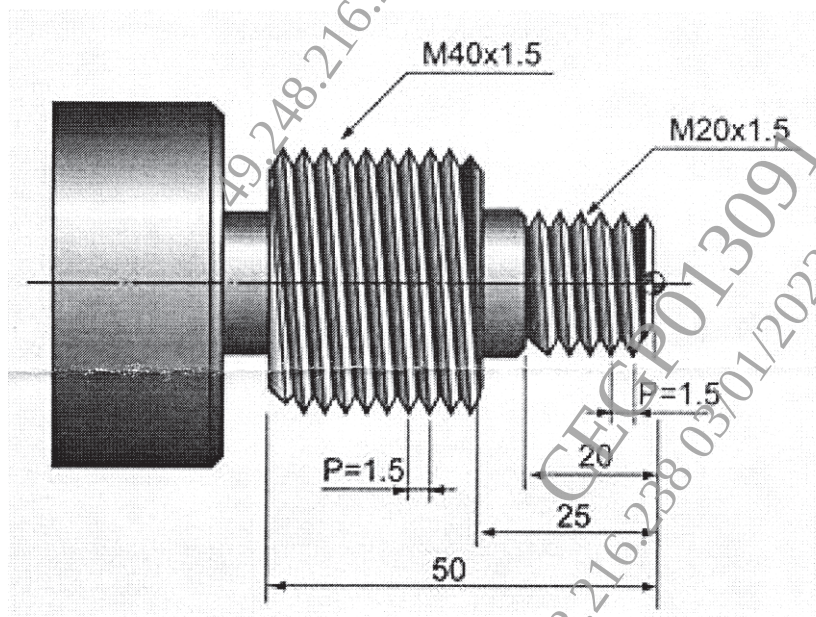
Figure 1: Qu 4

- Q5)** a) Write steps in FEA with neat sketch [6]
 b) Write CNC part programming for Milling for component as shown in figure 2 using concept of sub-programming with 1 mm step. Assume suitable cutting data. [12]



OR

- Q6)** a) Explain difference between canned cycle for roughing and finishing operations in CNC turning with neat sketch. [6]
 b) Write CNC part program for turned component as shown in figure 3 using canned cycles. Assume suitable cutting data. [12]



- Q7)** a) Write short note on Components of Product Life Cycle (PLM). [8]
b) Explain FDM with neat sketch. [8]

OR

- Q8)** a) Discuss: Classification of RP processes. [8]
b) What is Collaborative Engineering and briefly discuss role of Collaborative Engineering in current industry. [8]

- Q9)** a) Differentiate between HARD & SOFT Automation. [8]
b) State 'Laws of Robotics'. [8]

OR

- Q10)** a) Write short note on Flexible Manufacturing Systems (FMS). [8]
b) Discuss briefly: Group Technology. [8]

